

WEEK 8

Wireless Technologies

Computer Networking: A Top-Down Approach
Kurose & Ross, 6th Edition — Chapter 6 & 8

Wi-Fi (802.11)

WiFi Security

Bluetooth

Zigbee

Wireless Security

8.1 Introduction to Wi-Fi

IEEE 802.11 Wireless LANs

Textbook Ref: Chapter 6, pp. 513–527



- Wireless LANs have become one of the most important access network technologies in the Internet today
- Coverage: home, workplace, cafés, airports, universities, even street corners — ubiquitous connectivity
- IEEE 802.11 standards (Wi-Fi) dominate; defined by the IEEE 802.11 working group [IEEE 802.11 2012]
 - 802.11 operates in unlicensed radio frequency bands (2.4 GHz and 5 GHz)

Standards Evolution:

- 802.11b — 2.4 GHz, up to 11 Mbps, widely deployed legacy standard
- 802.11a — 5 GHz, up to 54 Mbps, shorter range due to higher frequency
- 802.11g — 2.4 GHz, up to 54 Mbps, backward compatible with 802.11b
- 802.11n — MIMO antennas, up to 450 Mbps, operates in both 2.4 GHz and 5 GHz

- Basic Service Set (BSS) — fundamental building block of 802.11 architecture
- Each BSS contains one or more wireless stations and a central Access Point (AP)
- AP connects the BSS to the wired infrastructure (router → Internet)
- Each 802.11 device has a unique 6-byte MAC address stored in firmware (NIC)

Service Set Identifier (SSID):

- Administrator assigns an SSID (1–2 word name) to each AP
- Administrator also assigns a channel number (1–11 in 2.4 GHz band)
- Channels 1, 6, and 11 are non-overlapping — key for deployment planning
- Passive Scanning: host listens for beacon frames from APs
- Active Scanning: host broadcasts a Probe Request; APs respond with Probe Response

Why Collision Avoidance (not Detection)?

- Wireless channels have much higher BER than wired — collisions very costly
- Hidden terminal problem: two stations out of range of each other but both reach AP
- Signal fading makes collision detection unreliable
- 802.11 uses CSMA/CA — sense first, then transmit with backoff
- Link-layer ACK scheme: receiver sends ACK after successful receipt

CSMA/CA Protocol Steps

- 1. If channel is IDLE: wait DIFS interval, then transmit
- 2. If channel is BUSY: choose random backoff (binary exponential backoff)
- 3. Count down backoff only while channel is idle
- 4. Transmit entire frame; wait for ACK
- 5. No ACK received → assume collision → double backoff, retry
- RTS/CTS optional: Reserve channel to solve hidden terminal problem

- 802.11 frame differs significantly from Ethernet — most notably it has FOUR address fields
- Frame Control (2 bytes) — type, subtype, flags (To DS, From DS, WEP bit, etc.)
- Duration (2 bytes) — channel reservation time (for virtual carrier sensing)

Address Fields (6 bytes each × 4):

- Address 1: MAC address of wireless station receiving this frame
- Address 2: MAC address of transmitting station or AP
- Address 3: MAC address of the router interface (gateway to Internet)
- Address 4: Used only in ad hoc mode (multi-hop mesh scenarios)
- Sequence Control (2 bytes) — sequence number for ARQ (duplicate detection)
- Data Payload — up to ~2,312 bytes (usually holds IP datagram)
- CRC (4 bytes) — error detection; critical due to high wireless BER

8.2 WiFi Security

Protecting 802.11 Wireless Networks

Textbook Ref: Chapter 8, pp. 626–636



Why Wireless Security is Critical

p. 626–627

- Radio waves carrying frames propagate far beyond building walls — any device in range can receive them
- An attacker can record wireless transmissions and replay them passively — no physical access required
- Original 802.11 specification contained serious security flaws — publicly acknowledged and widely exploited
- Public domain software can exploit original 802.11 security holes — making WEP-only networks essentially open

Core Security Requirements:

- Confidentiality — prevent eavesdropping on wireless frames
- Authentication — verify identity of connecting device/user
- Integrity — detect tampering with wireless frames in transit
- Access Control — restrict network access to authorized users

WEP: Wired Equivalent Privacy (Broken)

p. 627–630



WEP Design Goal

Originally designed to provide security equivalent to wired LAN. Uses symmetric RC4 stream cipher with a 40-bit or 104-bit shared key.



IV Problem

24-bit Initialization Vector (IV) prepended to key. Only $2^{24} = 16$ million possible IVs — IV reuse occurs after ~12,000 frames.



Known-Plaintext Attack

Attacker observes ciphertext and sends known content to AP. When IV repeats, attacker can recover the keystream and decrypt all traffic.



Fatal Flaws

IV transmitted in plaintext, key never refreshed, weak key scheduling in RC4. WEP is completely broken — should NEVER be used.

- 802.11i provides robust security for wireless LANs — uses separate authentication server (not just the AP)

Four Phases of 802.11i Operation:

- Phase 1 — Discovery: AP advertises its presence and supported authentication/encryption methods
- Phase 2 — Mutual Authentication & Master Key (MK) Generation:
 - Authentication between wireless client and Authentication Server (AS)
 - AP acts as relay only — keeps AP simple and low-cost
 - Uses EAP (Extensible Authentication Protocol) over RADIUS
 - EAP-TLS uses public key cryptography for mutual authentication
- Phase 3 — Pairwise Master Key (PMK): Derived from MK; shared by client and AP (not AS)
- Phase 4 — Temporal Key (TK) Generation: Per-session keys used for actual frame encryption
- Encryption: AES-based (strong) or enhanced WEP (TKIP) — AES strongly preferred

8.3 WiFi Exploitation

Common Wireless Attack Vectors

Textbook Ref: Chapter 8, pp. 627–636



- Understanding attacks is essential to building effective defenses

Passive Eavesdropping (Sniffing):

- Attacker places wireless adapter in promiscuous mode — captures all frames on the channel
- WEP-encrypted traffic can be cracked in minutes with tools like aircrack-ng

Evil Twin / Rogue AP Attack:

- Attacker deploys an AP with the same SSID as a legitimate network
- Victims connect to rogue AP — all traffic flows through attacker (MITM)

Deauthentication Attack:

- 802.11 management frames are unauthenticated — attacker sends fake deauth frames
- Disconnects legitimate clients; can force them to reconnect and capture 4-way handshake
- WPA2 Handshake Cracking — captured handshake is subject to offline dictionary attacks

8.4 Bluetooth Working

IEEE 802.15.1 Personal Area Networks

Textbook Ref: Chapter 6, pp. 551–553



- Bluetooth (IEEE 802.15.1) is designed for short-range, low-power cable replacement — a Wireless Personal Area Network (WPAN)

Piconet Architecture:

- One Master device controls up to 7 Active Slaves in a piconet
- Up to 255 additional Parked (inactive) devices can be registered but cannot communicate until activated
- All communication is coordinated by the master — slave-to-slave is not direct

Physical Layer:

- Operates in 2.4 GHz ISM (unlicensed) band — same band as 802.11b/g
- Uses frequency-hopping spread spectrum (FHSS) — hops among 79 channels, 1600 hops/sec
- FHSS provides robustness against interference and coexistence with Wi-Fi
- Typical range: 10 meters (Class 2) to 100 meters (Class 1 with amplifier)
- Data rates: 1 Mbps (v1.0), 3 Mbps (v2.0 + EDR), 24 Mbps (v3.0 + HS)

Protocol Layers

- Radio Layer — frequency hopping, modulation (GFSK, $\pi/4$ -DPSK)
- Baseband Layer — link control, packet format, piconet management
- LMP (Link Manager Protocol) — authentication, encryption, power control
- L2CAP — logical link control and adaptation; multiplexes upper protocols
- RFCOMM — serial port emulation (cable replacement)
- Profiles: A2DP (audio), HFP (headset), HID (keyboard/mouse), OBEX (files)

Key Operational Features

- Inquiry & Paging: Device discovery and connection setup process
- Time Division Duplex (TDD): Master and slave alternate transmission slots
- Power Modes: Active, Sniff, Hold, Park — optimize battery life
- Scatternet: Multiple piconets can overlap; a device can be master in one and slave in another
- Adaptive Frequency Hopping (AFH): avoids crowded channels dynamically

8.5 Zigbee Working

IEEE 802.15.4 Low-Power Mesh Networks

Textbook Ref: Chapter 6, pp. 553–555



- Zigbee targets lower-powered, lower-data-rate, lower-duty-cycle applications than Bluetooth — ideal for IoT sensors and smart home devices
- Not all applications need high bandwidth — sensor readings, on/off commands, status updates are very small

Key Specifications:

- Frequency: 2.4 GHz (global), 868 MHz (Europe), 915 MHz (Americas)
- Data Rate: 250 kbps at 2.4 GHz (much lower than Bluetooth/Wi-Fi)
- Range: 10–100 meters (mesh extends range dramatically)
- Power: designed for years of battery life — devices can sleep most of the time

Superframe Structure:

- Time is divided into recurring superframes, each starting with a beacon frame
- Active period: Contention Access Period (CAP) + Contention Free Period (CFP)
- Inactive period: coordinator and devices sleep — major power saving

Zigbee Network Types & Device Roles

p. 553–555



Full Function Device (FFD)

Can act as coordinator, router, or end device. Implements the full 802.15.4 standard. Can relay messages for other devices in mesh topology.



Reduced Function Device (RFD)

Simplified implementation. Can only communicate with FFDs (not other RFDs). Lower cost, lower power. Ideal for simple sensors and actuators.



Star Topology

All devices communicate through a single central coordinator (FFD). Simple, low cost. Limited range — all devices must reach coordinator directly.



Mesh/Cluster Tree Topology

FFDs route messages hop-by-hop across large areas. Self-healing — alternate routes used if a node fails. Enables city-scale sensor networks.

8.6 Bluetooth & Zigbee Security

Security Vulnerabilities & Protections

Textbook Ref: Chapter 6 & 8



Bluetooth Security Architecture

Ref: BT Core Spec 5.x

- Bluetooth security is built on pairing — establishing a shared secret key between devices

Classic Bluetooth Security Modes:

- Mode 1: No security — device accepts all connections without authentication
- Mode 2: Service-level security — security enforced per-service after connection
- Mode 3: Link-level security — authentication and encryption before connection established
- Mode 4 (BT 2.1+): Secure Simple Pairing (SSP) with Elliptic Curve Diffie-Hellman

Cryptographic Mechanisms:

- Authentication: Challenge-response using 128-bit link key
- Encryption: E0 stream cipher (128-bit key) applied to all air traffic
- Key Generation: PIN-based (legacy) or SSP (modern) pairing
- Common Attacks: Bluejacking, Bluesnarfing, Bluebugging, MITM on weak pairing

Bluetooth vs. Zigbee Security Comparison

Ref: IEEE 802.15.4/Zigbee
Spec

Bluetooth Security

- Pairing model: PIN (legacy) or Secure Simple Pairing
- Encryption: E0 stream cipher (Classic), AES-CCM (BLE)
- Authentication: Mutual via challenge-response (link key)
- Key hierarchy: Link Key → Encryption Key
- Vulnerabilities: Weak PINs, MITM on JustWorks pairing, BlueBorne exploits
- BLE: AES-128 CCM provides strong confidentiality and integrity

Zigbee Security

- Uses AES-128 encryption (CCM* mode) at MAC and network layers
- Three key types: Master Key, Trust Center Link Key, Network Key
- Trust Center: central authority manages key distribution
- Network Key: shared across the entire network for broadcast
- Vulnerabilities: Key sent in plaintext during join (early implementations), replay attacks if counter not checked
- Zigbee 3.0: mandatory security — fixed many earlier weaknesses

Week 8 Summary

8.1 Wi-Fi Intro

Ch.6 p.526

802.11 architecture (BSS/AP), channels, SSID, scanning

MAC Protocol

Ch.6 p.530

CSMA/CA, DIFS, backoff, RTS/CTS, 4-address frame format

8.2 WiFi Security

Ch.8 p.627

WEP flaws (IV reuse, RC4 weakness), 802.11i phases, EAP-TLS, AES

8.3 Exploitation

Ch.8 p.626

Sniffing, Evil Twin, Deauth attacks, WPA2 handshake capture

8.4 Bluetooth

Ch.6 p.551

IEEE 802.15.1, piconet, FHSS, 79 channels, 1600 hops/sec

8.5 Zigbee

Ch.6 p.553

IEEE 802.15.4, 250 kbps, superframe, FFD/RFD, mesh topology

8.6 BT/ZB Security

Ch.6/8

BT link keys, E0/AES, Zigbee AES-128 CCM*, Trust Center